

A Metric Counterexample: Nearly Isometric Embeddability Need Not Imply Almost Lipschitz Embeddability

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Source Problem

This packet addresses Problem 4 in Section 5 of:

Florent Baudier and Gilles Lancien, *Tight embeddability of proper and stable metric spaces*, arXiv:1503.06089.

The source problem, on PDF page 17, asks:

Exhibit two metric spaces X and Y such that X nearly isometrically embeds into Y , but X does not almost Lipschitz embed into Y .

Definitions Used

We use the definitions from the source paper. A metric space X almost Lipschitz embeds into Y if there are constants $r > 0$, $D \geq 1$, such that for every continuous $\varphi : [0, \infty) \rightarrow [0, 1)$ with $\varphi(0) = 0$ and $\varphi(t) > 0$ for $t > 0$, there is a map $f_\varphi : X \rightarrow Y$ satisfying

$$r d_X(x, y) \varphi(d_X(x, y)) \leq d_Y(f_\varphi(x), f_\varphi(y)) \leq D r d_X(x, y).$$

For nearly isometric embeddability, the admissible compressions $\rho \in \mathcal{P}$ are continuous functions with $\rho(t) = t$ on $[0, 1]$, $\rho(t) \leq t$ for $t \geq 1$, and $\rho(t)/t \rightarrow 0$. The admissible expansions $\omega \in \Omega$ satisfy $\omega(0) = 0$, $t \leq \omega(t)$ on $[0, 1]$, $\omega(t) = t$ for $t \geq 1$, and $\omega(t)/t \rightarrow \infty$ as $t \downarrow 0$. The space X nearly isometrically embeds into Y if for every such pair (ρ, ω) there is $f : X \rightarrow Y$ with

$$\rho(d_X(x, y)) \leq d_Y(f(x), f(y)) \leq \omega(d_X(x, y)).$$

Proof Intuition

The source definition of nearly isometric embeddability permits the embedding to depend on the requested sublinear compression ρ . We exploit this by building a target Y with one ray for each admissible ρ . The ρ -ray is metrically distorted by a sublinear function g_ρ that still dominates ρ , so the usual ray $\mathbb{N}_0$ can satisfy that particular nearly isometric request by entering that component.

The obstruction to almost Lipschitz embeddability is global. An almost Lipschitz embedding of the integer ray would become a bi-Lipschitz embedding after choosing a test function φ bounded below on all integer distances. But the bouquet target has no bi-Lipschitz ray. A Lipschitz image with linear lower growth must eventually be far from the common basepoint; at that height it cannot switch components, because switching costs the sum of the two heights. Once trapped in one component, distances grow only sublinearly, contradicting linear compression.

Auxiliary Concave Majorants

Lemma 1. *Let $\rho : [0, \infty) \rightarrow [0, \infty)$ be continuous, satisfy $\rho(t) = t$ on $[0, 1]$, $\rho(t) \leq t$ for $t \geq 1$, and $\rho(t)/t \rightarrow 0$ as $t \rightarrow \infty$. Then there is a continuous, nondecreasing, concave, unbounded function $g : [0, \infty) \rightarrow [0, \infty)$ such that*

$$g(0) = 0, \quad g(t) = t \quad (0 \leq t \leq 1), \quad \rho(t) \leq g(t) \leq t, \quad \frac{g(t)}{t} \rightarrow 0.$$

Consequently $d_g(m, n) = g(|m - n|)$ is a metric on $\mathbb{N}_0$.

Proof. Let $h(t) = t$ for $0 \leq t \leq 1$, and $h(t) = 1 + \log t$ for $t \geq 1$. Then h is continuous, nondecreasing, concave, unbounded, $h(t) \leq t$, and $h(t)/t \rightarrow 0$. Put $\rho_0(t) = \max\{\rho(t), h(t)\}$. Then $\rho_0(t) \leq t$, $\rho_0(t) = t$ on $[0, 1]$, and $\rho_0(t)/t \rightarrow 0$.

Choose a sequence $\varepsilon_j \downarrow 0$. Since $\rho_0(t)/t \rightarrow 0$, for each j the number

$$A_j = \sup_{t \geq 0} (\rho_0(t) - \varepsilon_j t)$$

is finite. Define

$$g(t) = \inf \left(\{t\} \cup \{A_j + \varepsilon_j t : j \geq 1\} \right).$$

Every affine function in the infimum dominates ρ_0 , and the function $t \mapsto t$ also dominates ρ_0 . Hence $g \geq \rho_0 \geq \rho$. Also $g \leq t$. Because $g \leq t$ and $g \geq \rho_0 = t$ on $[0, 1]$, we have $g(t) = t$ on $[0, 1]$.

The pointwise infimum of increasing affine functions is concave and nondecreasing. For each fixed j ,

$$\limsup_{t \rightarrow \infty} \frac{g(t)}{t} \leq \limsup_{t \rightarrow \infty} \frac{A_j + \varepsilon_j t}{t} = \varepsilon_j,$$

and therefore $g(t)/t \rightarrow 0$. Finally, $g \geq h$, so g is unbounded.

A nondecreasing concave function with $g(0) = 0$ is subadditive: concavity implies $g(s)/s \geq g(s+t)/(s+t)$ and $g(t)/t \geq g(s+t)/(s+t)$ for $s, t > 0$, whence $g(s) + g(t) \geq g(s+t)$. Thus $g(|m - n|)$ satisfies the triangle inequality on $\mathbb{N}_0$. $\square$

The Counterexample

Theorem 1. *There are metric spaces X and Y such that X nearly isometrically embeds into Y , but X does not almost Lipschitz embed into Y .*

Proof. Let $X = \mathbb{N}_0$ with its usual metric $d_X(m, n) = |m - n|$. For every admissible $\rho \in \mathcal{P}$, choose a function g_ρ supplied by Lemma 1. Let $Y_\rho = \mathbb{N}_0$ with metric

$$d_\rho(m, n) = g_\rho(|m - n|).$$

Now form the metric wedge Y of the family $(Y_\rho)_{\rho \in \mathcal{P}}$ by identifying all zero points as a common basepoint o . If $a \in Y_\rho$ and $b \in Y_\sigma$, set

$$d_Y(a, b) = \begin{cases} g_\rho(|a - b|), & \rho = \sigma, \\ g_\rho(a) + g_\sigma(b), & \rho \neq \sigma. \end{cases}$$

This is the usual wedge metric: inside each component we use d_ρ , and between different components we travel through o .

We first show that X nearly isometrically embeds into Y . Fix $\rho \in \mathcal{P}$ and $\omega \in \Omega$. Define $F_{\rho,\omega} : X \rightarrow Y$ by $F_{\rho,\omega}(n) = n$ in the ρ -component. If $m, n \in \mathbb{N}_0$ and $k = |m - n|$, then for $k = 0$ there is nothing to check. For $k \geq 1$, the defining properties of g_ρ and the fact that $\omega(k) = k$ give

$$\rho(k) \leq g_\rho(k) = d_Y(F_{\rho,\omega}(m), F_{\rho,\omega}(n)) \leq k = \omega(k).$$

Thus X nearly isometrically embeds into Y .

It remains to show that X does not almost Lipschitz embed into Y . We first prove that Y contains no bi-Lipschitz copy of the usual ray. Suppose, toward a contradiction, that $F : \mathbb{N}_0 \rightarrow Y$ and constants $c, C > 0$ satisfy

$$c|m - n| \leq d_Y(F(m), F(n)) \leq C|m - n| \quad (m, n \in \mathbb{N}_0).$$

Let $R_n = d_Y(F(n), o)$. Since

$$R_n \geq d_Y(F(n), F(0)) - R_0 \geq cn - R_0,$$

the heights R_n grow at least linearly. On the other hand, $d_Y(F(n+1), F(n)) \leq C$. If $F(n)$ and $F(n+1)$ lie in different components, then

$$d_Y(F(n+1), F(n)) = R_{n+1} + R_n,$$

which is eventually larger than C . Hence, for all sufficiently large n , consecutive points $F(n)$ and $F(n+1)$ lie in the same component. The tail of the sequence $F(n)$ is therefore contained in a single component Y_{ρ_0} .

Write $F(n) = a_n \in Y_{\rho_0}$ for all $n \geq N$. Since

$$g_{\rho_0}(|a_{n+1} - a_n|) = d_Y(F(n+1), F(n)) \leq C$$

and g_{ρ_0} is nondecreasing and unbounded, there is $B < \infty$ such that $|a_{n+1} - a_n| \leq B$ for all $n \geq N$. Therefore

$$|a_m - a_n| \leq B|m - n| \quad (m, n \geq N).$$

For $m, n \geq N$,

$$d_Y(F(m), F(n)) = g_{\rho_0}(|a_m - a_n|) \leq g_{\rho_0}(B|m - n|).$$

Dividing by $|m - n|$ and letting $|m - n| \rightarrow \infty$, the right-hand side has ratio tending to 0 because $g_{\rho_0}(t)/t \rightarrow 0$. This contradicts the lower bound $d_Y(F(m), F(n)) \geq c|m - n|$. Thus no bi-Lipschitz embedding $\mathbb{N}_0 \rightarrow Y$ exists.

Finally suppose that X almost Lipschitz embeds into Y . Take

$$\varphi(t) = \begin{cases} t/2, & 0 \leq t \leq 1, \\ 1/2, & t \geq 1. \end{cases}$$

This φ is admissible in the almost Lipschitz definition. Since all nonzero distances in $X = \mathbb{N}_0$ are at least 1, the corresponding map would satisfy

$$\frac{r}{2}|m - n| \leq d_Y(f_\varphi(m), f_\varphi(n)) \leq Dr|m - n| \quad (m \neq n),$$

which is a bi-Lipschitz embedding of the ray into Y . This contradicts the previous paragraph. Hence X does not almost Lipschitz embed into Y . $\square$

Verification Notes

The construction is intentionally nonseparable in general, since it uses one component for every admissible compression function ρ . The source problem only asks for metric spaces, so no separability or Banach structure is being claimed. The proof uses no external theorem beyond elementary facts about concave metric transforms and wedge metrics.

Bounded Novelty Check

On 2026-06-15, the run indexes were searched for 1503.06089, `nearly isometric`, and `almost Lipschitz`; no duplicate packet or attempt for this metric problem was found. Local parsed-source search for phrases such as `nearly isometrically embeds` and `does not almost Lipschitz` found the source paper but no separate answer. Web searches for exact phrases including "`nearly isometrically embeds`" "`almost Lipschitz`", "`Exhibit two metric spaces`" "`nearly isometrically`" "`almost Lipschitz`", and `site:arxiv.org` "`almost Lipschitz`" "`nearly isometric`" returned the source paper and no later answer. The novelty claim is therefore bounded by these searches.

References

- F. Baudier and G. Lancien, *Tight embeddability of proper and stable metric spaces*, arXiv:1503.06089.

Human Review: Nearly isometric does not imply almost Lipschitz

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct. The construction is a clever exploitation of the main difference between nearly isometric embeddability and almost Lipschitz embeddability: in the nearly isometric definition, the embedding is allowed to depend on the requested compression function ρ . The target space can therefore be made large enough to contain a separate component adapted to each admissible compression.

By contrast, the almost Lipschitz condition is more global. In the present example $X = \mathbb{N}_0$, all nonzero distances are at least 1. Choosing an admissible function φ which is bounded below by a positive constant on $[1, \infty)$ turns an almost Lipschitz embedding into an ordinary bi-Lipschitz embedding of the integer ray. Thus it is enough to construct a target Y which satisfies every individual nearly isometric compression request, but which contains no bi-Lipschitz copy of $\mathbb{N}_0$.

Construction

For each admissible compression ρ , the proof chooses a concave, nondecreasing, unbounded function g_ρ such that

$$\rho(t) \leq g_\rho(t) \leq t, \quad \frac{g_\rho(t)}{t} \rightarrow 0.$$

The ρ -component is a copy of $\mathbb{N}_0$ with metric

$$d_\rho(m, n) = g_\rho(|m - n|).$$

Then Y is the wedge, or bouquet, of all these components at their common basepoint.

This immediately gives nearly isometric embeddability. Given a requested pair (ρ, ω) , map $n \in \mathbb{N}_0$ to n inside the ρ -component. On integer distances $k \geq 1$, one has

$$\rho(k) \leq g_\rho(k) \leq k = \omega(k),$$

so the required inequalities hold.

No Bi-Lipschitz Ray

The obstruction to almost Lipschitz embeddability is that Y contains no bi-Lipschitz copy of the usual ray. Suppose, for contradiction, that

$$F : \mathbb{N}_0 \rightarrow Y$$

is bi-Lipschitz. Let

$$R_n = d_Y(F(n), o)$$

be the distance from the common basepoint. The lower Lipschitz estimate gives linear growth:

$$R_n \geq cn - R_0.$$

On the other hand, consecutive points satisfy

$$d_Y(F(n+1), F(n)) \leq C.$$

If $F(n)$ and $F(n+1)$ were in different bouquet components, their distance would be

$$R_n + R_{n+1},$$

which is eventually larger than C . Hence, after some point, consecutive terms must lie in the same component. Therefore the whole tail of the ray is trapped in a single component Y_{ρ_0} .

Inside this component, distances are measured by

$$g_{\rho_0}(|a_m - a_n|).$$

The upper Lipschitz bound on consecutive points implies that the underlying integer jumps $|a_{n+1} - a_n|$ are uniformly bounded. Thus

$$|a_m - a_n| \leq B|m - n|.$$

But g_{ρ_0} is sublinear, so

$$\frac{g_{\rho_0}(B|m - n|)}{|m - n|} \rightarrow 0.$$

This contradicts the lower linear bound required by a bi-Lipschitz embedding.

Review Outcome

Approve this packet as an apparently correct counterexample to the source problem. The key idea is exactly the dependence of the nearly isometric embedding on ρ : the bouquet can accommodate every requested sublinear compression separately, while the almost Lipschitz condition, in this discrete setting, would force a genuine bi-Lipschitz ray, which the bouquet is designed to exclude.